
ME 614: Compressible Flow and Gas Dynamics

Course Code	ME 614
Course Name	Compressible Flow and Gas Dynamics
L-T-P-C	2.5-0.5-0-3
Prerequisites	Fluid Mechanics, Thermodynamics, Conservation Laws
Intended for	MS/PhD
Distribution	Elective
Approval	3rd Ad-hoc Senate (Nov 2012)

Course Contents

- **Introduction**

Gas dynamics, review of basic mass, momentum and energy conservation laws for compressible flows, speed of sound, wave equation, regimes of mach number, shocks, wave propagation, sound speed, Mach number, isentropic flow, static and stagnation properties. [6 Lectures]

- **One Dimensional Flow**

Converging-diverging nozzles, shock waves, moving and reflected waves, blast waves, wind tunnels, supersonic engines [8 Lectures]

- **Two Dimensional Flow**

Oblique shock wave theory, conical oblique shock waves, Prandtl-Mayer expansion Fans, supersonic inlets and diffusers. [8 Lectures]

- **Compressible Pipe Flow**

Fanno-Line flow, Rayleigh pipe flow, natural gas flow in pipelines. [3 Lectures]

- **Compressible Potential Flow**

Method of characteristics, supersonics nozzle design. [6 Lectures]

Text and Reference Books

1. J.D. Anderson, **Modern Compressible Flow (With Historical Perspective)**, 2nd Edition, McGraw-Hill, 1990
2. S M Yahya, **Fundamentals of Compressible Flow**, 4th Edition, New Age International, 2010.
3. Shapiro, Ascher H., **Dynamics and thermodynamics of compressible fluid flow**, Vol 1 and Vol 2, John Wiley, 1953.